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# Residual-based a posteriori error estimate for a parareal approximation of the cardiac monodomain problem

Jad Dabaghi\*

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## Abstract

In this work, we propose a “parareal-in-time” approach over massively parallel computing environments to accelerate the numerical resolution of the monodomain problem devoted to simulate the electrical activity within the heart tissue. From a mathematical standpoint, the monodomain problem is modeled by a nonlinear parabolic partial differential equation coupled with an ordinary differential equation whose numerical resolution is often out of reach. The parareal algorithm is based on two different propagators: a low-cost “coarse” propagator based on a very cheap Galerkin scheme and a “fine” propagator based on a high-fidelity and thus expensive resolution. The fine propagator is performed in parallel so that the whole computational cost of the simulation is dramatically reduced. Furthermore, at each parareal iteration, we derive an *a posteriori* error estimate on the error between the exact solution at the continuous level and the approximate solution given by the parareal scheme. In particular our approach enables to distinguish the different sources of the error: the spatial error and the parareal-in-time error. In a set of numerical experiments we compare our parareal strategy with a standard resolution. In particular, we show that for a large number of processors, our parareal approach significantly reduces the computational cost of the simulation while preserving its accuracy.

**Keywords:** Monodomain model; Reaction-diffusion system; Parareal-in-time algorithm; Galerkin schemes; *A posteriori* error estimates; Adaptivity; Cardiac electrophysiology.

## 1 Introduction

Cardiovascular diseases are the main cause of death worldwide. Studying them represents a significant challenge for the scientific community in life science and biomedical engineering. From the mathematical point of view, various models have been developed to simulate accurately the electrical activity of the heart. We refer to [1, 2] for a global overview. Perhaps, one of the most accurate descriptions of the electrical cardiac behavior is given by the so-called “bidomain model” [3, 4, 2]. Established by Schmitt [5] and first formally developed by Tung [6], the bidomain formulation consists of two partial differential equations (PDEs) modeling the intra- and extracellular potentials coupled with a system of ordinary differential equations (ODEs) describing the ion currents across the membrane. Assuming that anisotropy ratios between intracellular and extracellular media are equal, we obtain a simplification of the bidomain formulation known as the monodomain model. From a mathematical standpoint, the monodomain formulation consists of a nonlinear parabolic PDE describing the transmembrane voltage, which is defined as the difference between the intra- and extracellular potentials, coupled with a system of ODEs representing the ionic dynamics, see also [3, 4, 7] and the references therein. Unfortunately, deriving analytical solutions for such models is often out of reach and numerical simulations appear to be in general the only viable approach to approximate the solution. Among the wide range of numerical methods for solving PDEs, we mention for instance the traditional finite element method [8, 9, 10], the finite volume method [11, 12, 13], the discontinuous Galerkin method [14, 15, 16] and the Hybrid High Order method [17, 18]. Some of these methods have been successfully employed in the context of cardiac electrophysiology. The literature is substantial in this topic and we

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mention for instance [19, 20] for finite element discretizations. Bendahmane et al. [21, 22] examined finite volume schemes for the monodomain and bidomain models, proving existence and uniqueness of the discrete solution. Chehade et al. [23] proposed a TPFA scheme for the microscopic bidomain model with error analysis and Coudière et al. [24] investigated a whole analysis of 3D finite volume scheme for the monodomain model including stability, convergence, and error estimates. Note that some recent contributions provide a posteriori error estimates for the monodomain problem [25, 26]. In particular, in [26], an adaptive version of the Newton algorithm based on equilibrated flux reconstructions [27, 28] is formulated in order to localize and distinguish the different source of the error and to reduce the number of iterations within the simulation. However, real time simulation for evolutive-in-time problems require important computational effort. In this context, parallelizing the treatment of the time variable appears to provide an effective compromise between computational efficiency and accuracy, while also alleviating memory requirements. The key idea behind the time parallelization is to decompose the time direction into “slices” where each interval can be handled in parallel. In this respect, several strategies have been proposed in order to efficiently cope with the parallelization in time. Perhaps, the first study dates back to the pioneering work of Nievergelt [29] where the Multiple Shooting method is introduced to solve in parallel an ordinary differential equation (ODE). While a classic resolution employs a sequential procedure, this approach proposes to consider that the set of all solutions to the ODE are the unknowns of a larger nonlinear problem. The benefits of such approach is to reduce the computational cost of the underlying nonlinear resolution by carrying out some computations in a parallel fashion. Next, Chartier and Philippe [30] extended the work of Nievergelt [29] to systems of ODE’s. The parareal algorithm [31] is structurally close to the Multiple Shooting method. One main advantage of this method, as shown in [31], is that it approximates the solution later in time, before having fully accurate approximations at earlier times, while preserving the global accuracy of the iterative process. It involves two propagators  $F$  and  $G$  that approximately integrate a given system of partial differential equations. The propagator  $F$  is a fine, accurate and thus expensive propagator, which approximates the exact solution with high accuracy; on the contrary, the propagator  $G$  is a coarse propagator, which is a less accurate approximation of the exact solution and much less expensive than  $F$ . A prediction term is first computed thanks to the coarse propagator  $G$ . Then, a correction step involving both propagators is performed in parallel. These solvers can, e.g., be based on different time steps ( $\delta t$  for  $F$  being typically much smaller than  $\Delta t$  for  $G$ ), but the model that  $G$  approximates may also be a simplified version of the underlying set of partial differential equations describing the model. Let  $T_0 = 0 < T_1 < \dots < T_N = T$  be a sequence of times. For the sake of simplicity, we choose here  $T_n = n\Delta T$  for some appropriate time interval  $\Delta T$ . The parareal algorithm constructs a sequence  $\mathbf{u}_k := (u_k^n)_{1 \leq n \leq N}$  such that, for each iteration  $k \geq 0$ ,  $u_k^n$  is an approximation of the analytical solution  $u$  of some PDE at time step  $n$  ( $u^n = u(n\Delta T)$ ). For the iteration  $k = 0$ , the initial approximation is obtained at each time step  $n$  using the coarse propagator  $G$  over a propagation length of fixed size  $\Delta T$  (we denote by  $G_{\Delta T}$  such a coarse evolution over a time window of size  $\Delta T$ ) :

$$\forall n \geq 0, \quad u_{k=0}^{n+1} := G_{\Delta T}(u_{k=0}^n), \quad (1.1)$$

where  $u_{k=0}^0 := u^0$ . Next, we perform a prediction, followed by a correction iteration

$$\forall n \geq 0, \quad u_{k+1}^{n+1} := \underbrace{G_{\Delta T}(u_{k+1}^n)}_{\text{prediction}} + \underbrace{[F_{\Delta T}(u_k^n) - G_{\Delta T}(u_k^n)]}_{\text{correction}} \quad (1.2)$$

where  $u_{k+1}^0 = u^0$ . Note that the coarse solver  $G$  is sequential, whereas the fine solver  $F$  computes at the end of each step  $k$  the corrections in parallel. When the algorithm converges,  $k \rightarrow +\infty$ , equation (1.2) yields  $u_{k+1}^{n+1} = F_{\Delta T}(u_k^n)$  and thus the final approximation is achieved by the accuracy of the fine propagator  $F$  with a weaker restitution clock time. More precisely, the reduction factor tends to the value of  $\frac{N}{k}$ . An attractive feature of this algorithm is that both fine and coarse propagator are treated as black-boxes and can encompass explicit or implicit methods as well as arbitrary direct or iterative linear or nonlinear solvers. The parareal algorithm has been successfully applied in several applications. We mention for instance reservoir simulations [32], pricing in american options [33] and kinetic transport problems to [34, 35, 36]. In particular in [36] is developed a hybrid version of the parareal algorithm where a deterministic coarse propagator is coupled with a stochastic fine propagator. Several theoretical convergence results of the parareal algorithm can be found in [31, 37, 38, 39]. Note also that a recent contribution [40] explores the ability of the parareal algorithm to converge without the coarse propagator. In particular, the authors show that for parabolic

problems with Dirichlet boundary condition, the use of a coarse propagator is not necessary to achieve satisfying convergence rate.

In this work, we are interested in constructing a parareal-in-time algorithm for accelerating the numerical resolution of the monodomain problem in electrocardiology. We also provide at each parareal iteration an *a posteriori* error estimate to have a guaranteed upper bound on the error between the analytical solution and the numerical solution. Several strategies have been developed in the past concerning a posteriori error estimates for partial differential equations. We mention for instance the residual based a posteriori error estimates [41, 42, 43, 44], the approach by equilibrated flux reconstructions [45, 46, 28, 27], or hierarchical estimates [47, 48, 49]. In this work we employ residual based *a posteriori* error estimates that present attractive features such as low numerical cost and smart location of the error. To the best of our knowledge, coupling a *a posteriori* error analysis with a parareal scheme for mathematical formulations in cardiac electrophysiology has never been presented before. A gap still remains in this topic and we structure the paper as follows.

In Section 2, we present the monodomain problem and its weak formulation. Next, in Section 3, we present an abstract framework dedicated to discretize the weak formulation. Furthermore, in Section 4 we explicit the whole parareal algorithm including its different stages. Next, in Section 5, we provide some classical discretization methods that could use within the parareal procedure. Then, Section 6 is devoted to the *a posteriori* analysis following the approach of the residual-based estimators. A heuristical distinction of the error components is also given. Finally, Section 7 reports some numerical experiments, assessing the strengths of our approach.

## 2 Model problem and setting

In this section, we set up the notation and describe in detail the monodomain model.

### 2.1 Monodomain model and setting

We consider a bounded, open, polygonal, connected domain  $\Omega \subset \mathbb{R}^d$ ,  $d = 2, 3$ , with Lipschitz boundary  $\partial\Omega$ . We denote by  $T > 0$  the final simulation time, and we set  $\Omega_T := (0, T) \times \Omega$ . Let  $\mathcal{D}(\Omega)$  be the space of  $C^\infty$  functions with compact support on  $\Omega$  and  $\mathcal{D}'(\Omega)$  its dual version. Let  $H^1(\Omega)$  be the space of  $L^2$  functions on the domain  $\Omega$  which admit a weak gradient in  $L^2(\Omega)$  and  $H_0^1(\Omega)$  its zero-trace subspace. The duality pairing between any Sobolev space  $V$  and its dual space  $V'$  is denoted by  $\langle \cdot, \cdot \rangle_{V', V}$ . The standard notations  $\nabla$  and  $\nabla \cdot$  are used respectively for the weak gradient and divergence operators. For a nonempty set  $\mathcal{O}$  of  $\mathbb{R}^2$ , we denote its Lebesgue measure by  $|\mathcal{O}|$  and the  $L^2(\mathcal{O})$  scalar product by  $(u, v)_{\mathcal{O}} := \int_{\mathcal{O}} uv \, dx$  for  $u, v \in L^2(\mathcal{O})$ . We also use the following notations:  $\|v\|_{\mathcal{O}}^2 := (v, v)_{\mathcal{O}}$ , and  $\|\nabla v\|_{\mathcal{O}}^2 := (\nabla v, \nabla v)_{\mathcal{O}}$ .

We consider the monodomain model:

$$\begin{aligned} \partial_t v - \nabla \cdot (\Lambda \nabla v) + f(v, w) &= 0 && \text{in } \Omega_T, \\ \partial_t w + g(v, w) &= 0 && \text{in } \Omega_T, \\ \Lambda \nabla v \cdot \mathbf{n} &= 0 && \text{on } (0, T) \times \partial\Omega, \\ v|_{t=0} &= v^0, \quad w|_{t=0} = w^0 && \text{in } \Omega. \end{aligned} \tag{2.1}$$

In System (2.1), the unknown variables are the transmembrane potential  $v(t, \mathbf{x}) \in \mathbb{R}$ , and the ionic variable  $w(t, \mathbf{x}) \in \mathbb{R}$  (gating variable, concentration, etc) which is an auxiliary function characterizing the underlying transfer of ions across the membrane in the considered medium. The first line of (2.1) describes the balance of current in the heart muscle where  $\partial_t v$  is the capacitive current,  $\nabla \cdot (\Lambda \nabla v)$  is the ohmic current, and  $f(v, w)$  is the ionic current induced by the cellular activation. In this model, the anisotropic characteristics of the tissue are given by the tensor  $\Lambda(\mathbf{x}) \in \mathbb{R}^{d \times d}$ . The second line of (2.1) describes the dynamic of the activation and inhibition variables for the ionic channels and  $g(v, w)$  is a function representing the ionic activity. In an isolated heart, no current flows out of the heart, as expressed by the boundary condition (third line of (2.1)) where  $\mathbf{n}$  is the outward unit normal vector to  $\partial\Omega$ . Finally, the last line of (2.1) corresponds to the initial conditions for the functions  $v$  and  $w$ .

**Assumption 2.1** (Assumptions on the conductivity tensor). *Assuming an anisotropic medium, the conductivity is represented by the tensor  $\Lambda(\mathbf{x})$  (with coefficients in  $L^\infty(\Omega)$ ) which is a symmetric, bounded, and uniformly positive definite on  $\Omega$ . It means that*

$$\exists (m_0, M_0) \in \mathbb{R}_+^* \times \mathbb{R}_+^* \text{ s.t. } \forall \boldsymbol{\xi} \in \mathbb{R}^d, 0 < m_0 |\boldsymbol{\xi}|^2 \leq \Lambda \boldsymbol{\xi} \cdot \boldsymbol{\xi} \leq M_0 |\boldsymbol{\xi}|^2.$$

**Assumption 2.2** (Assumptions on the ionic functions (see [50])). *The functions  $f$  and  $g$  are affine with respect to  $w$  and verify*

$$f(v, w) = f_1(v) + f_2(v)w, \quad g(v, w) = g_1(v) + g_2w$$

where  $f_1 : \mathbb{R} \rightarrow \mathbb{R}$ ,  $f_2 : \mathbb{R} \rightarrow \mathbb{R}$ , and  $g_1 : \mathbb{R} \rightarrow \mathbb{R}$  are continuous functions, and  $g_2 \in \mathbb{R}$ . We also assume that there exist positive constants  $c_j \geq 0$ ,  $j = 1 \dots 6$  such that  $\forall v \in \mathbb{R}$ , for all  $2 \leq p \leq 6$ ,

$$|f_1(v)| \leq c_1 + c_2|v|^{p-1}, \quad |f_2(v)| \leq c_3 + c_4|v|^{p/2-1}, \quad |g_1(v)| \leq c_5 + c_6|v|^{p/2}$$

Under these assumptions, the mappings  $f : L^p(\Omega) \times L^2(\Omega) \rightarrow L^{p'}(\Omega)$  and  $g : L^p(\Omega) \times L^2(\Omega) \rightarrow L^2(\Omega)$  are well defined and satisfy (see [50, Lemma 25]):

$$\begin{aligned} \|f(v, w)\|_{L^{p'}(\Omega)} &\leq A_1|\Omega|^{1/p'} + A_2\|v\|_{L^p(\Omega)}^{p/p'} + A_3\|w\|_{L^2(\Omega)}^{2/p'}, \\ \|g(v, w)\|_{L^2(\Omega)} &\leq B_1|\Omega|^{\frac{1}{2}} + B_2\|v\|_{L^p(\Omega)}^{p/2} + B_3\|w\|_{L^2(\Omega)}. \end{aligned} \quad (2.2)$$

In inequalities (2.2) the terms  $A_1, A_2, A_3$ , and  $B_1, B_2, B_3$  stand for positive constants that depend on the parameters  $p$  and  $c_j$  for  $j = 1 \dots 6$ .

**Assumption 2.3** (minoration). *The functions  $f$  and  $g$  satisfy the minoration (2.3) (see [51]). There exist  $C_0 \in \mathbb{R}$ ,  $C_1 > 0$ ,  $C_2 > 0$  and  $r > 0$  such that for any  $(v, w) \in \mathbb{R}^2$ ,*

$$C_0 + C_1|v|^p + C_2|w|^2 \leq \frac{r}{C} f(v, w)v - g(v, w)w, \quad (2.3)$$

with  $C$  the cellular membrane capacitance.

**Remark 2.4.** *In the mathematical analysis of monodomain equations, we distinguish several paths to define ionic currents. In general two models are employed : the phenomenological and physiological models. The first one simplifies the underlying mechanisms and focuses on reproducing characteristic behaviors (such as ventricular dynamics) at the macroscopic scale. In terms of computational cost, these models are competitive but unfortunately do not capture the entire physical processes. Perhaps, one of the most well-known phenomenological model is the FitzHugh–Nagumo model [52, 53] that satisfies Assumptions 2.2. We mention other phenomenological ionic models: the Roger-McCulloch model [54], the Aliev-Panfilov model [55], and the Mitchell-Schaeffer model [56], which may be considered more general than the FitzHugh–Nagumo model but exhibit some mathematical difficulties. The physiological models, based on the cell membrane formulation developed by Hodgkin-Huxley [57] for nerve fibers, are more realistic and capture several physical phenomena at different scales such as the mechanical response of the medium or the microscopic behavior of chemical species. However, the mathematical models developed in this context are systems of nonlinear and parabolic PDEs coupled with multiple ODEs. Numerical simulation would involve many unknowns and an unaffordable computational cost.*

## 2.2 Weak formulation

Before we define our Galerkin discretization scheme, we start by stating the weak formulation associated with Problem (2.1).

**Definition 2.5.** *Given  $f \in L^2(0, T; (H^1(\Omega))')$ ,  $g \in L^2(0, T; L^2(\Omega))$ ,  $v_0 \in L^2(\Omega)$ ,  $w_0 \in L^2(\Omega)$  and  $\Lambda \in L^\infty(\Omega)$ , a weak solution of problem (2.1) is a vector  $\mathbf{U} := (v, w)$  of functions such that  $v \in L^2(0, T; H^1(\Omega))$ ,  $\partial_t v \in L^2(0, T; (H^1(\Omega))')$  and  $w \in \mathcal{C}(0, T; L^2(\Omega))$  verifying  $\forall \varphi \in H^1(\Omega)$  and  $\forall \phi \in L^2(\Omega)$*

$$\begin{aligned} \frac{d}{dt} \langle v(t), \varphi \rangle_{(H^1(\Omega))', H^1(\Omega)} + (\Lambda \nabla v(t), \nabla \varphi)_\Omega &= - \langle f(v(t), w(t)), \varphi \rangle_{(H^1(\Omega))', H^1(\Omega)} \\ (\partial_t w(t), \phi)_\Omega &= - (g(v(t), w(t)), \phi)_\Omega. \end{aligned} \quad (2.4)$$

Regarding the monodomain problem (2.1), the existence of its weak solution is given in the following theorem.

**Remark 2.6** (Well-posedness of Monodomain Model). *Assume that the conditions listed in Assumptions 2.1 and 2.2 hold. Then, the monodomain problem (2.1) possesses a unique weak solution in the sense of Definition 2.4. The proof of such result is standard and relies on semigroup arguments presented in [58, Lemma 31] or on the Faedo-Galerkin procedure (see [59] and the references therein).*

### 3 Discretization methods and propagators

In this section, we present the numerical discretization of problem (2.4) to define the coarse and fine propagators involved within the parareal scheme. In particular, we specify the coarse and fine grid for the discretization in time. Several spatial propagators are presented for the coarse resolution (the high-fidelity resolution follows the same rules with a smaller time step).

#### 3.1 Time discretization

For the coarse time discretization, we introduce a division of the interval  $[0, T]$  into subintervals  $I_{n+1} := [t_{n-1}, t_n]$ ,  $1 \leq n \leq N_t$ , such that  $0 = t_0 < t_1 < \dots < t_{N_t} = T$ . The time steps are denoted by  $\Delta t_n = t_n - t_{n-1}$ , with  $n = 1, \dots, N_t$ . For a function of time  $v$  with sufficient regularity, we denote  $v^n := v(t_n)$ ,  $0 \leq n \leq N_t$ , and, for  $1 \leq n \leq N_t$ , we define the backward differencing operator

$$\partial_t v^n := \frac{1}{\Delta t_n} (v^n - v^{n-1}).$$

Concerning the fine propagator, we consider a subdivision of the interval  $[0, T]$  into subintervals  $[\tilde{t}_{n-1}, \tilde{t}_n]$   $1 \leq n \leq N^*$  such that  $0 = \tilde{t}_0 < \tilde{t}_1 < \dots < \tilde{t}_{N^*} = T$ . The time steps are denoted by  $\delta t_n = \tilde{t}_n - \tilde{t}_{n-1}$ ,  $n = 1 \dots N^*$ . Note that here,  $\delta t_n \ll \Delta t_n$ .

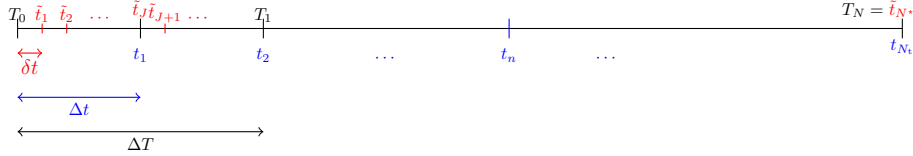


Figure 1: Subdivision of the time interval  $[0, T]$ .  $\Delta T$  : time window,  $\Delta t_n$  : time step associated to the coarse propagator,  $\delta t_n$  : time step associated to the fine propagator.

#### 3.2 Space Discretization

Let  $V_h$  be a finite dimensional Hilbert space equipped with the scalar product  $(\cdot, \cdot)_{V_h}$ . We also denote by  $m > 1$  the cardinality of the space  $V_h$ . We do not specify any specific hypotheses on the space  $V_h$ : the discretization can be conforming ( $V_h \subset H^1(\Omega)$ ) or not conforming ( $V_h \not\subset H^1(\Omega)$ ). We introduce a bilinear continuous coercive form  $a_h : V_h \times V_h \rightarrow \mathbb{R}$ . Given  $v_h^0 \in V_h$  and  $w_h^0 \in V_h$ , the discrete formulation corresponding to problem (2.4) consists, for each  $1 \leq n \leq N_t$ , in searching  $(v_h^n, w_h^n) \in V_h \times V_h$  such that for all  $(\varphi_h, \phi_h) \in V_h \times V_h$ ,

$$\begin{aligned} \frac{1}{\Delta t_n} (v_h^n - v_h^{n-1}, \varphi_h)_{V_h} + a_h(v_h^n, \varphi_h) &= \langle f(v_h^n, w_h^n), \varphi_h \rangle_{V_h', V_h} \quad \forall \varphi_h \in V_h \\ \frac{1}{\Delta t_n} (w_h^n - w_h^{n-1}, \phi_h)_{V_h} &= -\langle g(v_h^n, w_h^n), \phi_h \rangle_{V_h', V_h} \quad \forall \phi_h \in V_h. \end{aligned} \quad (3.1)$$

Let  $(\psi_i)_{1 \leq i \leq m}$  be a basis of the space  $V_h$ . We have the decomposition

$$v_h^n = \sum_{i=1}^n [\mathbf{X}_{1h}^n]_i \psi_i, \quad w_h^n = \sum_{i=1}^m [\mathbf{X}_{2h}^n]_i \psi_i, \quad \text{where} \quad \mathbf{X}_h^n := (\mathbf{X}_{1h}^n, \mathbf{X}_{2h}^n) \in \mathbb{R}^{2m}, \quad (3.2)$$

Setting  $\varphi_h = \phi_h = \psi_j, \forall 1 \leq j \leq m$  and denoting by  $\mathbb{A} \in \mathbb{R}^{m,m}$  and  $\mathbb{M} \in \mathbb{R}^{m,m}$  the matrices defined by

$$\mathbb{A}_{i,j} := a_h(\psi_i, \psi_j), \quad \mathbb{M}_{i,j} := (\psi_i, \psi_j)_\Omega, \quad \forall 1 \leq i, j \leq m \quad (3.3)$$

and by  $\mathbf{F} \in \mathbb{R}^m$  and  $\mathbf{G} \in \mathbb{R}^m$  the vectors defined by

$$\mathbf{F}_i := (f(v_h^n, w_h^n), \psi_i)_\Omega \quad \mathbf{G}_i := (g(v_h^n, w_h^n), \psi_i)_\Omega \quad \forall 1 \leq i \leq m \quad (3.4)$$

the system (3.1) reads

$$\begin{pmatrix} \frac{1}{\Delta t_n} \mathbb{M} + \mathbb{A} & 0 \mathbb{I}_d \\ 0 \mathbb{I}_d & \frac{1}{\Delta t_n} \mathbb{M} \end{pmatrix} \begin{pmatrix} \mathbf{X}_{1h}^n \\ \mathbf{X}_{2h}^n \end{pmatrix} - \begin{pmatrix} \frac{1}{\Delta t_n} \mathbb{M} & 0 \mathbb{I}_d \\ 0 \mathbb{I}_d & -\frac{1}{\Delta t_n} \mathbb{M} \end{pmatrix} \begin{pmatrix} \mathbf{X}_{1h}^{n-1} \\ \mathbf{X}_{2h}^{n-1} \end{pmatrix} = \begin{pmatrix} \mathbf{F}(\mathbf{X}_h^n) \\ -\mathbf{G}(\mathbf{X}_h^n) \end{pmatrix}. \quad (3.5)$$

Note that in (3.5) the matrix  $\mathbb{I}_d \in \mathbb{R}^{m,m}$  is the identity matrix. We thus obtain the following nonlinear system of algebraic equations : Given  $\mathbf{X}_h^{n-1} = [\mathbf{X}_{1h}^{n-1}, \mathbf{X}_{2h}^{n-1}] \in \mathbb{R}^{2m}$  find  $\mathbf{X}_h^n \in \mathbb{R}^{2m}$  solution to

$$\mathcal{S}(\mathbf{X}_h^n) = 0. \quad (3.6)$$

### 3.3 Newton method

In this section we propose a Newton linearization procedure to solve the nonlinear system (3.6) is defined as follows. Let an initial guess  $\mathbf{X}_h^{n,0} \in \mathbb{R}^{2m}$  be given, typically  $\mathbf{X}_h^{n,0} = \mathbf{X}_h^{n-1}$ . Here,  $\mathbf{X}_h^{n-1}$  is the last iterate from the previous time step. At linearization step  $l \geq 1$ , one looks for  $\mathbf{d}_h^{n,l} \in \mathbb{R}^{2m}$  such that

$$\mathbb{J}_\mathcal{S}(\mathbf{X}_h^{n,l-1}) \mathbf{d}_h^{n,l} = -\mathcal{S}(\mathbf{X}_h^{n,l-1}). \quad (3.7)$$

In (3.7) the term  $\mathbb{J}_\mathcal{S}(\mathbf{X}_h^{n,l-1}) \in \mathbb{R}^{2m,2m}$  is the Jacobian matrix of the function  $\mathcal{S}$  at point  $\mathbf{X}_h^{n,l-1}$ . Once the Newton direction  $\mathbf{d}_h^{n,l} \in \mathbb{R}^{2m}$  is computed we obtain the Newton solution  $\mathbf{X}_h^{n,l}$  by

$$\mathbf{X}_h^{n,l} = \mathbf{d}_h^{n,l} + \mathbf{X}_h^{n,l-1}$$

**Remark 3.1.** Concerning the stopping criterion of the Newton solver, we choose a tolerance  $\varepsilon > 0$  close to the machine precision and stop the linearization procedure when the error satisfies  $\|\mathcal{S}(\mathbf{X}_h^{n,l})\|_2 \leq \varepsilon$ . In this case we set  $\mathbf{X}_h^n = \mathbf{X}_h^{n,l}$ . Note that in general, the linear system (3.7) is not solved exactly especially when the number of degrees of freedom is large. We would rather employ at each Newton step  $k \geq 1$  an iterative algebraic solver. This enters in the wide range of inexact Newton algorithms [60, 61, 62] which is a common approach to speed it up.

## 4 The parareal algorithm

In Section 3.1–3.3 we presented a generic space-time discretization of the model problem (2.1). The parareal algorithm that we use in the sequel is based on two propagators : a low cost coarse propagator denoted by  $\mathcal{G}$  where the time step  $\Delta t$  is large enough, and an expensive fine propagator denoted by  $\mathcal{F}$  with an associated time step  $\delta t \ll \Delta t$ . These propagators are computed thanks to Section 3 where  $\mathcal{S}$  is replaced by  $\mathcal{F}_{\delta t}$  or by  $\mathcal{G}_{\Delta t}$ .



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**Algorithm 1** The parareal algorithm

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**Initialization** Choose an initial vector  $\mathbf{X}_h^0 \in \mathbb{R}^{2m}$  and compute a coarse approximation  $\mathbf{X}_{h,k=0}^{n+1} \in \mathbb{R}^{2m}$  of the numerical unknown  $\mathbf{X}_h^{n+1}$  at each time observable  $(n+1)\Delta T$ ,  $0 \leq n \leq N-1$ , by the coarse propagator

$$\mathbf{X}_{h,k=0}^{n+1} = \mathcal{G}_{\Delta T}(\mathbf{X}_{h,k=0}^n) \quad \text{where} \quad \mathbf{X}_{h,k=0}^0 = \mathbf{X}_h^0.$$

**For**  $k = 1 : \bar{k}$

**For**  $n = 0 : N-1$  (in parallel)

        Compute the correction term:

$$\mathcal{F}_{\Delta T}(\mathbf{X}_{h,k}^n) - \mathcal{G}_{\Delta T}(\mathbf{X}_{h,k}^n). \quad (4.1)$$

**end**

**For**  $n = 0 : N-1$

        Compute the prediction term  $\mathcal{G}_{\Delta T}(\mathbf{X}_{h,k+1}^n)$  and next the parareal solution at the time observable  $(n+1)\Delta T$ :

$$\mathbf{X}_{h,k+1}^{n+1} = \mathcal{G}_{\Delta T}(\mathbf{X}_{h,k+1}^n) + [\mathcal{F}_{\Delta T}(\mathbf{X}_{h,k}^n) - \mathcal{G}_{\Delta T}(\mathbf{X}_{h,k}^n)]. \quad (4.2)$$

**end**

**For**  $n = 0 : N-1$  (loop for parareal update)

        Check the stopping criterion:

$$\sup |\mathbf{X}_{h,k+1}^{n+1} - \mathbf{X}_{h,k}^{n+1}| \leq \varepsilon_k. \quad (4.3)$$

        If satisfied, set  $\mathbf{X}_h^{n+1} = \mathbf{X}_{h,k}^{n+1}$ . If not, set  $k := k+1$  and go back to the loop indexed by  $k$ .

**end**

**end**

---

**Remark 4.1.** In Algorithm 1 is presented the parareal scheme for the resolution of the monodomain formulation. In the first stage (Initialization), some coarse approximations are computed at each time step  $1 \leq n \leq N$ . Obviously these approximations are not faithful but are cheap in terms of computational cost. The coarse propagator  $\mathcal{G}_{\Delta T}$  correspond to the use of some discretization scheme as presented in Section 3 (see also Section 5 for classical applications). Note that these coarse approximations are computed once from the beginning. Next, stage (4.1) corresponds to the computation of the correction terms in parallel. More precisely, we employ  $N$  processors and for each one is computed a correction term. Next, in stage (4.2) we compute sequentially the parareal update by computing first a prediction term. The last stage (4.3) corresponds to the stopping criterion of the parareal scheme. If for a current parareal window the stopping criterion (4.3) is satisfied for all time steps one can stop the parareal algorithm. Note that the computational cost of this Algorithm is mainly based on the use of the fine solver.

In the sequel we present some classical propagators that could be employed in the context of the parareal algorithm. However, the a posteriori analysis developed in this work is restricted to the finite element method.

## 5 Application to classical numerical methods

**The finite element method** Let  $\mathcal{T}_h$  be a conforming mesh of  $\Omega$ , i.e.  $\mathcal{T}_h$  is a set of triangles verifying

$$\bigcup_{K \in \mathcal{T}_h} \bar{K} = \bar{\Omega}$$

where the intersection of the closure of two elements of  $\mathcal{T}_h$  is either an empty set, a vertex, or an edge. For  $K \in \mathcal{T}_h$ ,  $h_K$  denotes the diameter of a triangle  $K$ . Then, we define the mesh diameter by  $h := \max_{K \in \mathcal{T}_h} h_K$ .



We also assume that the mesh  $\mathcal{T}_h$  is shape regular in the sense that there exists a constant  $\mathcal{K}_{\mathcal{T}} > 0$  such that,  $\max_{K \in \mathcal{T}_h} \frac{h_K}{\rho_K} \leq \mathcal{K}_{\mathcal{T}}$ , where  $\rho_K$  is the diameter of the largest ball inscribed in  $K$ . Note that in two space dimensions, shape regularity means that the smallest angle in the mesh  $\mathcal{T}_h$  is uniformly bounded away from zero. Thus, we do not consider so-called anisotropic meshes.

The set of vertices of  $\mathcal{T}_h$  is denoted by  $\mathcal{V}_h$  and is partitioned into the interior vertices  $\mathcal{V}_h^{\text{int}}$  and the boundary vertices  $\mathcal{V}_h^{\text{ext}}$ . We denote by  $\mathcal{N}_h^{\text{int}}$  the number of interior vertices and by  $\mathcal{N}_h^{\text{ext}}$  the number of boundary vertices. Then,  $\mathcal{N}_h$  denotes the total number of vertices. The vertices of an element  $K \in \mathcal{T}_h$  are collected in the set  $\mathcal{V}_K$ . Furthermore, for  $\mathbf{a} \in \mathcal{V}_h$ , let the patch  $\omega_h^{\mathbf{a}} \subset \Omega$  be the domain made up of the elements of  $\mathcal{T}_h$  that share  $\mathbf{a}$ . The vector  $\mathbf{n}_{\omega_h^{\mathbf{a}}}$  represents its outward unit normal. In the same way,  $\omega_K \subset \Omega$  is the domain composed by the neighbors of  $K$ , more precisely  $\omega_K = \bigcup_{\mathcal{E}_K \cap \mathcal{E}_{K'} \neq \emptyset} K'$ ; and  $\omega_{\sigma} = \bigcup_{\sigma \in \mathcal{E}_K} K$ .

We also denote by  $\mathcal{V}_d^p$  the set of the Lagrange nodes (points  $\mathbf{x}_l$ ) of the space  $X_h^p$  and by  $\mathcal{N}_d^p$  its cardinality. The internal Lagrange nodes are collected in the set  $\mathcal{V}_d^{p,\text{int}}$  (with  $\mathcal{N}_d^{p,\text{int}}$  its cardinality) and the external ones are collected in the set  $\mathcal{V}_d^{p,\text{ext}}$  (with  $\mathcal{N}_d^{p,\text{ext}}$  its cardinality). In a same way, the set of edges of the mesh is denoted by  $\mathcal{E}_h$ . The internal respectively external edges are collected in the set  $\mathcal{E}_h^{\text{int}}$ , respectively  $\mathcal{E}_h^{\text{ext}}$ . Finally, for a sufficiently regular function  $v$ , we define the jump of  $v$  on  $\sigma = K \cap L \in \mathcal{E}_h^{\text{int}}$  as  $[[v]]_{\sigma} := (v|_K)_{\sigma} - (v|_L)_{\sigma}$ .

We use the discrete conforming space of piecewise polynomial functions

$$X_h^p := \{v_h \in C^0(\bar{\Omega}); v_h|_K \in \mathbb{P}_p(K) \forall K \in \mathcal{T}_h\} \subset H^1(\Omega)$$

where  $\mathbb{P}_p(K)$  stands for the set of polynomials of total degree less than or equal to  $p \geq 1$  on the element  $K \in \mathcal{T}_h$ . The Lagrange basis functions of  $X_h^p$  are denoted by  $(\psi_{h,\mathbf{x}_l})_{1 \leq l \leq \mathcal{N}_d^p}$  for  $\mathbf{x}_l \in \mathcal{V}_d^p$ . We recall that  $\psi_{h,\mathbf{x}_l}(\mathbf{x}_l) = 1$  and  $\psi_{h,\mathbf{x}_l}(\mathbf{x}_{l'}) = 0$  for all  $(\mathbf{x}_{l'})_{1 \leq l' \neq l \leq \mathcal{N}_d^p} \in \mathcal{V}_d^p$ . In the particular case  $p = 1$ , the set  $\mathcal{V}_d^1$  coincides with the mesh vertices  $\mathcal{V}_h$  and the Lagrange basis functions are the "hat" basis functions and are denoted by  $\psi_{h,\mathbf{a}}, \mathbf{a} \in \mathcal{V}_h$ .

**Remark 5.1.** We observe that in the context of finite elements, our discretization fits our abstract framework of Section 3 where  $V_h = X_h^p$  and  $a_h : X_h^p \times X_h^p \rightarrow \mathbb{R}$  is the bilinear coercive form defined by  $a_h(z_{1h}, z_{2h}) := (\Lambda \nabla z_{1h}, \nabla z_{2h})$ . Furthermore,  $m = \mathcal{N}_d^p$ .

Let  $p \geq 1$ . Given  $(v_h^0, w_h^0) \in X_h^p \times X_h^p$ , a discrete formulation corresponding to (2.4) consists in searching for all  $1 \leq n \leq N_t$ ,  $\mathbf{U}_h^n := (v_h^n, w_h^n) \in X_h^p \times X_h^p$  such that

$$\frac{1}{\Delta t_n} (v_h^n - v_h^{n-1}, \varphi_h)_{\Omega} + (\Lambda \nabla v_h^n, \nabla \varphi_h)_{\Omega} = - (f(v_h^n, w_h^n), \varphi_h)_{\Omega} \quad (5.1a)$$

$$\frac{1}{\Delta t_n} (w_h^n - w_h^{n-1}, \phi_h)_{\Omega} = - (g(v_h^n, w_h^n), \phi_h)_{\Omega}, \quad (5.1b)$$

for all  $\varphi_h$  and  $\phi_h$  in  $X_h^p$ . Furthermore, we want to express the discrete problem (3.1) under an algebraic form. Let  $p \geq 1$  and let  $(v_h^n, w_h^n)$  be the solution of problem (3.1). First, we decompose  $v_h^n$  and  $w_h^n$  in the Lagrange basis as follows

$$v_h^n = \sum_{l=1}^{\mathcal{N}_d^p} [\mathbf{X}_{1h}^n]_l \psi_{h,\mathbf{x}_l}, \quad w_h^n = \sum_{l=1}^{\mathcal{N}_d^p} [\mathbf{X}_{2h}^n]_l \psi_{h,\mathbf{x}_l}, \quad \text{where } \mathbf{X}_h^n := (\mathbf{X}_{1h}^n, \mathbf{X}_{2h}^n) \in \mathbb{R}^{2\mathcal{N}_d^p}, \quad (5.2)$$

with  $(\psi_{h,\mathbf{x}_l})_{1 \leq l \leq \mathcal{N}_d^p}$  the Lagrange basis for  $p \geq 2$  and  $[\mathbf{X}_{1h}^n]_l := v_h^n(\mathbf{x}_l)$  and  $[\mathbf{X}_{2h}^n]_l := w_h^n(\mathbf{x}_l)$ . We denote by  $\mathbb{A} \in \mathbb{R}^{\mathcal{N}_d^p, \mathcal{N}_d^p}$  respectively  $\mathbb{M} \in \mathbb{R}^{\mathcal{N}_d^p, \mathcal{N}_d^p}$ , the stiffness matrix, respectively the mass matrix defined by

$$\mathbb{A}_{l,m} := (\Lambda \nabla \psi_{h,\mathbf{x}_l}, \nabla \psi_{h,\mathbf{x}_m})_{\Omega}, \quad \text{and } \mathbb{M}_{l,m} := (\psi_{h,\mathbf{x}_l}, \psi_{h,\mathbf{x}_m})_{\Omega} \quad \forall 1 \leq l, m \leq \mathcal{N}_d^p. \quad (5.3)$$

Here also the matrices in (5.3) correspond to those of (3.3). Taking  $\varphi_h = \psi_{h,\mathbf{x}_m}$  in the first line of (3.1) and  $\phi_h = \psi_{h,\mathbf{x}_m}$  in the second line of (3.1) for all  $1 \leq m \leq \mathcal{N}_d^p$  we get

$$\begin{aligned} \frac{1}{\Delta t_n} \mathbb{M} \mathbf{X}_{1h}^n + \mathbb{A} \mathbf{X}_{1h}^n &= \frac{1}{\Delta t_n} \mathbb{M} \mathbf{X}_{1h}^{n-1} - \mathbf{F}(\mathbf{X}_h^n) \\ \frac{1}{\Delta t_n} \mathbb{M} \mathbf{X}_{2h}^n &= \frac{1}{\Delta t_n} \mathbb{M} \mathbf{X}_{2h}^{n-1} - \mathbf{G}(\mathbf{X}_h^n) \end{aligned} \quad (5.4)$$

where the nonlinear functions  $\mathbf{F} : \mathbb{R}^{2\mathcal{N}_d^p} \rightarrow \mathbb{R}^{\mathcal{N}_d^p}$  and  $\mathbf{G} : \mathbb{R}^{2\mathcal{N}_d^p} \rightarrow \mathbb{R}^{\mathcal{N}_d^p}$  are defined by  $\forall l \leq \mathcal{N}_d^p$

$$\begin{aligned} [\mathbf{F}(\mathbf{X}_h^n)]_l &= (f(v_h^n, w_h^n), \psi_{h,\mathbf{x}_l})_\Omega = (f_1(v_h^n) + f_2(v_h^n)w_h^n, \psi_{h,\mathbf{x}_l})_\Omega, \\ [\mathbf{G}(\mathbf{X}_h^n)]_l &= (g(v_h^n, w_h^n), \psi_{h,\mathbf{x}_l})_\Omega = (g_1(v_h^n) + g_2(v_h^n)w_h^n, \psi_{h,\mathbf{x}_l})_\Omega. \end{aligned} \quad (5.5)$$

To compute numerically (5.5) we employ the Ionic Current Interpolation (ICI) method [63, 64]. This method is widely adopted in the cardiac modeling community and relies on interpolating the ionic current after its evaluation at the Lagrange nodes. We get

$$(f_1(v_h^n), \psi_{h,\mathbf{x}_l})_\Omega = \sum_{m=1}^{\mathcal{N}_d^p} f_1([\mathbf{X}_{1h}^n]_m) (\psi_{h,\mathbf{x}_m}, \psi_{h,\mathbf{x}_l})_\Omega = \mathbb{M} \mathbf{F}_1 \quad (5.6)$$

where  $[\mathbf{F}_1]_l = f_1([\mathbf{X}_{1h}^n]_l)$  and the matrix  $\mathbb{M}$  is defined by (5.3). In the same spirit

$$\begin{aligned} (f_2(v_h^n)w_h^n, \psi_{h,\mathbf{x}_l})_\Omega &= \sum_{m=1}^{\mathcal{N}_d^p} f_2([\mathbf{X}_{1h}^n]_m) \sum_{l'=1}^{\mathcal{N}_d^p} [\mathbf{X}_{2h}^n]_{l'} (\psi_{h,\mathbf{x}_m} \psi_{h,\mathbf{x}_{l'}}, \psi_{h,\mathbf{x}_l})_\Omega \\ &= \mathbf{F}_2^T \tilde{\mathbb{M}} \mathbf{X}_{2h} \end{aligned} \quad (5.7)$$

where  $[\mathbf{F}_2]_l = [f_2([\mathbf{X}_{1h}^n]_l)]$  and  $\tilde{\mathbb{M}} \in \mathbb{R}^{\mathcal{N}_d^p \times \mathcal{N}_d^p}$  is the matrix defined by

$$\tilde{\mathbb{M}}_{l',m} = (\psi_{h,\mathbf{x}_m} \psi_{h,\mathbf{x}_{l'}}, \psi_{h,\mathbf{x}_l})_\Omega. \quad (5.8)$$

Finally, we obtain the compact nonlinear system of algebraic equations: given  $\mathbf{X}_h^{n-1} \in \mathbb{R}^{2\mathcal{N}_d^p}$  find  $\mathbf{X}_h^n \in \mathbb{R}^{2\mathcal{N}_d^p}$  solution to

$$\begin{cases} \mathcal{S}_1(\mathbf{X}_h^n) = 0 \\ \mathcal{S}_2(\mathbf{X}_h^n) = 0 \end{cases} \quad (5.9)$$

where  $\mathcal{S}_1 : \mathbb{R}^{2\mathcal{N}_d^p} \rightarrow \mathbb{R}^{\mathcal{N}_d^p}$  and  $\mathcal{S}_2 : \mathbb{R}^{2\mathcal{N}_d^p} \rightarrow \mathbb{R}^{\mathcal{N}_d^p}$  are given by (5.4). Introducing the operator  $\mathcal{S} : \mathbb{R}^{2\mathcal{N}_d^p} \rightarrow \mathbb{R}^{2\mathcal{N}_d^p}$  such that

$$\mathcal{S}(\mathbf{X}_h^n) := [\mathcal{S}_1(\mathbf{X}_h^n), \mathcal{S}_2(\mathbf{X}_h^n)] \quad (5.10)$$

we have to solve the nonlinear problem  $\mathcal{S}(\mathbf{X}_h^n) = 0$  thanks to the Newton method presented above in Section 3.3.

**Remark 5.2.** Observe that the linearization procedure of Section 3.3 can be particularly expensive since it requires to solve a system containing  $2\mathcal{N}_d^p$  unknowns. To speed up the numerical resolution, we employ the Schur complement who enables to solve a  $\mathcal{N}_d^p \times \mathcal{N}_d^p$  system of algebraic equations. More precisely  $\mathbf{X}_{1h}^{n,k} \in \mathbb{R}^{\mathcal{N}_d^p}$  is solution to

$$\left( \frac{1}{\Delta t_n} \mathbb{M} + \mathbb{S} + \mathbb{J}_{\tilde{\mathbf{F}}}(\mathbf{X}_{1h}^{n,k-1}) + \frac{a\lambda}{b + \frac{1}{\Delta t_n}} \mathbb{M} \right) \mathbf{X}_{1h}^{n,k} = \mathbf{B}_1^{n,k-1} - \frac{\lambda}{b + \frac{1}{\Delta t_n}} \mathbf{B}_2^{n,k-1} \quad (5.11)$$

Once  $\mathbf{X}_{1h}^{n,k}$  is computed, the vector  $\mathbf{X}_{2h}^{n,k} \in \mathbb{R}^{\mathcal{N}_d^p}$  is obtained as follows

$$\mathbf{X}_{2h}^{n,k} = \frac{1}{b + \frac{1}{\Delta t_n}} \mathbb{M}^{-1} \mathbf{B}_1^{n,k-1} + \frac{a}{b + \frac{1}{\Delta t_n}} \mathbf{X}_{1h}^{n,k}. \quad (5.12)$$

**The cell-centered finite volume method** In the context of the cell-centered finite volume method, we also assume that the family  $\mathcal{T}_h$  is superadmissible in the sense that for all cells  $K \in \mathcal{T}_h$  there exists a point  $\mathbf{x}_K \in K$  (the cell center) and for all edges  $\sigma \in \mathcal{E}_h$  there exists a point  $\mathbf{x}_\sigma \in \sigma$  (the edge center) such that, for all edges  $\sigma \in \mathcal{E}_K$ , the line segment joining  $\mathbf{x}_K$  with  $\mathbf{x}_\sigma$  is orthogonal to  $\sigma$  (see [65]). For an interior edge  $\sigma \in \mathcal{E}_h^{\text{int}}$  shared by two elements  $K$  and  $L$  (denoted in the sequel by  $\sigma = K \cap L$ ), we define the distance between these elements  $d_{KL} := \text{dist}(\mathbf{x}_K, \mathbf{x}_L)$ . We denote by  $K\sigma$  the mirror element of the element  $K$  such that  $\sigma = K \cap K\sigma$ . Furthermore, the total number of elements composing the mesh is denoted by  $\mathcal{N}_{\text{sp}}$ . Using

the cell-centered finite volume method, the unknowns of the model are discretized using one value per cell:  $\forall 1 \leq n \leq N_t$  we let

$$\mathbf{X}_h^n = (\mathbf{X}_K^n)_{K \in \mathcal{T}_h} \in \mathbb{R}^{2\mathcal{N}_{\text{sp}}} \quad \text{where} \quad \mathbf{X}_K^n = [X_{1K}^n, X_{2K}^n] \in \mathbb{R}^2 \quad (5.13)$$

where  $X_{1K}^n$  and  $X_{2K}^n$  are respectively the discrete elementwise unknowns approximating the values of  $v$  and  $w$  in the element  $K \in \mathcal{T}_h$ . To approximate the space gradient we use  $(\nabla v \cdot \mathbf{n}_{K,\sigma}, 1)_\sigma = |\sigma| \frac{v_{K\sigma} - v_K}{d_{KL}}$  if  $\sigma \in \mathcal{E}_K^{\text{int}}$ ,  $\bar{\sigma} = \overline{K} \cap \overline{K\sigma}$ . First, we discretize the diffusion equation in (2.1). Let  $K \in \mathcal{T}_h$ . By integration over the element  $K$ , and using the Green formula we obtain

$$\frac{|K|}{\Delta t} (v_K^n - v_K^{n-1}) - \sum_{\sigma \in \mathcal{E}_K} ((\Lambda \nabla v^n) \cdot \mathbf{n}_{K,\sigma}, 1)_\sigma + |K|f(v_K^n, w_K^n) = 0.$$

For any edge  $\sigma \in \mathcal{E}_h^{\text{int}}$ , such that  $\sigma = \overline{K} \cap \overline{L}$  we employ the following harmonic averaging formula [66] to get an approximation  $\mathcal{F}_{K,\sigma}^n$  of  $((\Lambda \nabla v^n) \cdot \mathbf{n}_{K,\sigma}, 1)_\sigma$ :

$$\mathcal{F}_{K,\sigma}^n := |\sigma| \frac{\Lambda_K \Lambda_L}{\Lambda_K d_{K\sigma} + \Lambda_L d_{L\sigma}} (v_L^n - v_K^n) \quad \text{with} \quad \Lambda_K = \mathbf{n}_{K,\sigma}^T \Lambda \mathbf{n}_{K,\sigma}.$$

In a same way we discretize the ODE equation as follows. Let  $K \in \mathcal{T}_h$ . By integration over the element  $K \in \mathcal{T}_h$  we obtain

$$\frac{|K|}{\Delta t} (w_K^n - w_K^{n-1}) + g(v_K^n, w_K^n) = 0. \quad (5.14)$$

Then, the finite volume scheme corresponding to (2.1) reads: for all  $1 \leq n \leq N_t$ , find  $\mathbf{X}_h^n \in \mathbb{R}^{2\mathcal{N}_{\text{sp}}}$  such that for all  $K \in \mathcal{T}_h$

$$\begin{aligned} \mathcal{H}_{1K}^n(\mathbf{X}_h^n) &= 0 \\ \mathcal{H}_{2K}^n(\mathbf{X}_h^n) &= 0 \end{aligned} \quad (5.15)$$

where the nonlinear functions  $\mathcal{H}_{1K}^n : \mathbb{R}^{2\mathcal{N}_{\text{sp}}} \rightarrow \mathbb{R}$  and  $\mathcal{H}_{2K}^n : \mathbb{R}^{2\mathcal{N}_{\text{sp}}} \rightarrow \mathbb{R}$  are defined by

$$\begin{aligned} \mathcal{H}_{1K}^n(\mathbf{X}_h^n) &= \frac{|K|}{\Delta t} (v_K^n - v_K^{n-1}) - \sum_{\sigma \in \mathcal{E}_K} \mathcal{F}_{K,\sigma}^n + |K|f(v_K^n, w_K^n) \\ \mathcal{H}_{2K}^n(\mathbf{X}_h^n) &= \frac{|K|}{\Delta t} (w_K^n - w_K^{n-1}) + g(v_K^n, w_K^n). \end{aligned} \quad (5.16)$$

Note that the system (5.15) is local. Its global version reads as:  $\mathcal{H}^n(\mathbf{X}_h^n) = 0$  where  $\mathcal{H}^n : \mathbb{R}^{2\mathcal{N}_{\text{sp}}} \rightarrow \mathbb{R}^{2\mathcal{N}_{\text{sp}}}$ . This nonlinear system is solved thanks to the Newton procedure presented above.

**Remark 5.3.** *We have considered the finite element method and the cell-centered finite volume method as propagators within the parareal framework. It is worth emphasizing, however, that the parareal algorithm is not restricted to these spatial discretization methods, and other numerical approaches can be employed as propagators, such as the discontinuous Galerkin method [67, 15] or the hybrid high-order method [17, 18]. In practice, the finite element method is among the most commonly used choices for the propagator.*

In Section 6, we present the a posteriori analysis of our model in the case of the finite element method. In particular we employ the approach of residual based a posteriori error estimates. The goal is to construct an upper bound on the error that depends only on the approximate parareal solution.

## 6 A posteriori error analysis for finite elements

### 6.1 Preamble

In Section 4 we have presented the parareal algorithm giving an approximation  $\mathbf{X}_{h,k}^n = [\mathbf{X}_{h,k}^{1,n}, \mathbf{X}_{h,k}^{2,n}] \in \mathbb{R}^{2m}$  of  $\mathbf{X}_h^n$  where  $1 \leq n \leq N_t$  is the time step and  $1 \leq k \leq \bar{k}$  is the parareal iteration. We are interested by deriving an a posteriori error estimate between the exact solution and the parareal solution. For the sake

of simplicity, we assume that the Newton solver has converged within each time iteration so that the index  $l$  is discarded. The functional representation of the vectors  $\mathbf{X}_{h,k}^{1,n}$  and  $\mathbf{X}_{h,k}^{2,n}$ , denoted by  $v_h^{n,k}$  and  $w_h^{n,k}$  are given by (5.2) for  $k \geq 1$ . Obviously,  $(v_h^{n,k}, w_h^{n,k}) \in X_h^p \times X_h^p \quad \forall 1 \leq n \leq N_t$ .

Next, we associate to the functions in space  $v_h^{n,k} \in X_h^p$ , and  $w_h^{n,k} \in X_h^p$ ,  $1 \leq n \leq N_t$ , their space-time representation  $v_{h\tau}^k$ , respectively  $w_{h\tau}^k$  as follows:

$$\begin{aligned} v_{h\tau}^k|_{I_n} &:= \frac{v_h^{n,k} - v_h^{n-1,k}}{\Delta t_n} (t - t^n) + v_h^{n,k} \quad \forall 1 \leq n \leq N_t, \\ w_{h\tau}^k|_{I_n} &:= \frac{w_h^{n,k} - w_h^{n-1,k}}{\Delta t_n} (t - t^n) + w_h^{n,k} \quad \forall 1 \leq n \leq N_t. \end{aligned} \quad (6.1)$$

Note that this construction ensures that  $v_{h\tau}^k$  and  $w_{h\tau}^k$  are piecewise affine and continuous in time, so that  $\partial_t v_{h\tau}^k \in L^2(0, T; (H^1(\Omega))') + L^{r/(r-1)}(\Omega_T)$  and  $\partial_t w_{h\tau}^k \in L^2(0, T; L^2(\Omega))$ . For each unknown, we note

$$v_{h\tau}^{n,k} := v_{h\tau}^k|_{I_n}, \quad w_{h\tau}^{n,k} := w_{h\tau}^k|_{I_n}.$$

Note that consequently

$$\partial_t v_{h\tau}^{n,k}|_{I_n} = \frac{1}{\Delta t_n} (v_h^{n,k} - v_h^{n-1,k}), \quad \partial_t w_{h\tau}^{n,k}|_{I_n} = \frac{1}{\Delta t_n} (w_h^{n,k} - w_h^{n-1,k}).$$

In the sequel, we will employ the Cl  ment interpolation operator. We recall it for the sake of clarity and we refer to [68, 10] for a complete description. This operator is denoted by  $\mathcal{I}_h : L^2(\Omega) \rightarrow H_0^1(\Omega)$  and is defined by

$$\mathcal{I}_h(v) = \sum_{\mathbf{x}_l \in \mathcal{V}_d^p} \left( \frac{1}{|\omega_{\mathbf{x}_l}|} \int_{\omega_{\mathbf{x}_l}} v(x) dx \right) \psi_{h,\mathbf{x}_l} \quad v \in L^2(\Omega).$$

This operator satisfies the following local error estimates  $\forall \hat{w} \in H_0^1(\Omega)$  and  $\forall K \in \mathcal{T}_h$

$$\|\hat{w} - \mathcal{I}_h(\hat{w})\|_K \leq c_A h_K \|\hat{w}\|_{H^1(\omega_K)} \quad (6.2)$$

and

$$\|\hat{w} - \mathcal{I}_h(\hat{w})\|_{L^2(\partial K)} \leq c_B h_K^{\frac{1}{2}} \|\hat{w}\|_{H^1(\omega_\sigma)}$$

where  $c_A > 0$  and  $c_B > 0$  are positive constants.

## 6.2 Residual of the equations

We denote by  $X_n$  the restriction of the energy space  $X := L^2(0, T; H^1(\Omega))$  to the time interval  $I_n$ , i.e.  $X_n := L^2(I_n, H^1(\Omega))$ . We also denote by  $Y_n$  the restriction of the energy space  $Y := L^2(0, T; L^2(\Omega))$  to the time interval  $I_n$ , i.e.  $Y_n := L^2(I_n, L^2(\Omega))$ . Then, we can introduce the following Sobolev space

$$Z_n := \{v \in X_n; \partial_t v \in X'_n := L^2(I_n, (H^1(\Omega))')\}$$

which is the restriction of the energy space

$$Z := \{v \in X; \partial_t v \in X' := L^2(0, T; (H^1(\Omega))')\}$$

to the time interval  $I_n$ .

For  $v \in X$  the space-time energy norm on  $X$  is given by

$$\|v\|_X := \left\{ \int_0^T \|v\|_\Omega^2(t) dt + \int_0^T \left\| \Lambda^{\frac{1}{2}} \nabla v \right\|_\Omega^2(t) dt \right\}^{\frac{1}{2}}.$$

We augment the energy norm by the dual norm of the time derivative, forming a norm on the space  $Z$ . For  $v \in Z$  we set

$$\|v\|_Z := \|v\|_X + \|\partial_t v\|_{X'}$$

where

$$\|\partial_t v\|_{X'} = \left\{ \int_0^T \|\partial_t v\|_{(H^1(\Omega))'}^2(t) dt \right\}^{\frac{1}{2}}.$$

Here we recall that

$$\|\partial_t v\|_{(H^1(\Omega))'} := \sup_{\varphi \in H^1(\Omega), \|\varphi\|_{H^1(\Omega)}=1} \langle \partial_t v, \varphi \rangle = \sup_{\varphi \in H^1(\Omega)} \frac{\langle \partial_t v, \varphi \rangle}{\|\varphi\|_{H^1(\Omega)}}.$$

**Definition 6.1.** *The residuals associated to (2.4) are defined by*

$$\begin{aligned} & \langle \mathcal{R}_1(v_{h\tau}^k), \varphi \rangle_{X', X} \\ &= \int_0^T \{ -(\partial_t v_{h\tau}^k, \varphi)_\Omega - (\Lambda \nabla v_{h\tau}^k, \nabla \varphi)_\Omega - \langle f(v_{h\tau}^k, w_{h\tau}^k), \varphi \rangle \} (t) dt \end{aligned} \quad (6.3)$$

$$\langle \mathcal{R}_2(w_{h\tau}^k), \psi \rangle_{Y, Y} = - \int_0^T \{ (\partial_t w_{h\tau}^k, \psi)_\Omega - (g(v_{h\tau}^k, w_{h\tau}^k), \psi)_\Omega \} (t) dt \quad (6.4)$$

for  $\varphi \in X$  and  $\psi \in Y$ .

The dual norm of the residuals is naturally defined by

$$\begin{aligned} \|\mathcal{R}_1(v_{h\tau}^k)\|_{X'} &= \sup_{\varphi \in X} \frac{|\langle \mathcal{R}_1(v_{h\tau}^k), \varphi \rangle_{X', X}|}{\|\varphi\|_X} \\ \|\mathcal{R}_2(w_{h\tau}^k)\|_Y &= \sup_{\psi \in Y} \frac{|\langle \mathcal{R}_2(w_{h\tau}^k), \psi \rangle_{Y, Y}|}{\|\psi\|_Y}. \end{aligned} \quad (6.5)$$

The following proposition links the dual norm of the residual with the energy norm. The proof follows the lines of [26, Proposition 4.7].

**Proposition 6.2.** *Let  $(v, w) \in X \times Y$  be the solution of (2.4) and let  $(v_{h\tau}^k, w_{h\tau}^k) \in X_h^p \times X_h^p$  be the approximate solution. Consider the conditions listed in Assumption 2.2. Then,*

$$\begin{aligned} & \|v - v_{h\tau}^k\|_X^2 + \|\partial_t(v - v_{h\tau}^k)\|_Y^2 + \|w - w_{h\tau}^k\|_Y^2 + \|\partial_t(w - w_{h\tau}^k)\|_Y^2 \\ & \leq C_\star \left( \|\mathcal{R}_1(v_{h\tau}^k)\|_Y^2 + \|\mathcal{R}_2(w_{h\tau}^k)\|_Y^2 \right. \\ & \quad \left. + \|(v - v_{h\tau}^k)(\cdot, 0)\|_\Omega^2 + \alpha_4 \|(w - w_{h\tau}^k)(\cdot, 0)\|_\Omega^2 \right) \end{aligned} \quad (6.6)$$

where  $C_\star$  is a generic constant that depends on  $\Omega$ ,  $f$ ,  $g$  and  $T$ .

### 6.3 A posteriori error estimates

In this section, for any polynomial degree  $p \geq 1$ , we derive an a posteriori error estimate on the error between the exact solution  $(v, w)$  given by the weak formulation (2.4) and the approximate solution  $(v_h^{n,k}, w_h^{n,k})$  given by (5.2) which is valid at any time step  $1 \leq n \leq N_t$  and any parareal iteration  $k \geq 1$ .

**Theorem 6.3.** *Let  $(v, w)$  be the solution of the continuous problem (2.1). Let  $(v_{h\tau}^k, w_{h\tau}^k)$  be the approximate solution stemming from the parareal scheme at each time step  $1 \leq n \leq N_t$  and each parareal iteration  $1 \leq k \leq \bar{k}$ . Let finally  $K \in \mathcal{T}_h$  and define the element residual estimators by*

$$\begin{aligned} \eta_{R_1, K}^{n,k}(t) &:= \left\| -\partial_t v_{h\tau}^k - f(v_{h\tau}^k, w_{h\tau}^k) + \nabla \cdot (\Lambda \nabla v_{h\tau}^k) \right\|_K(t) \\ \eta_{R_2, K}^{n,k}(t) &:= \left\| \partial_t w_{h\tau}^{n,k} + g(v_{h\tau}^{n,k}, w_{h\tau}^{n,k}) \right\|_K(t) \end{aligned} \quad (6.7)$$

the face residual estimator by

$$\eta_{R_3, \sigma}^{n,k}(t) := h_\sigma^{\frac{1}{2}} \left\| \llbracket \Lambda \nabla v_{h\tau}^k \rrbracket_\sigma \cdot \mathbf{n}_\sigma \right\|_\sigma(t) \quad (6.8)$$

and the initial condition estimators by

$$\eta_{\text{IC},1,K}^k := \|(v - v_{h\tau}^k)(\cdot, 0)\|_K, \quad \eta_{\text{IC},2,K}^k := \sqrt{\alpha_4} \|(w - w_{h\tau}^k)(\cdot, 0)\|_K. \quad (6.9)$$

Then, there exists a generic constant  $C_{\text{rel}}$  only depending on the space dimension  $d$ , the mesh shape regularity parameter  $\mathcal{K}_{\mathcal{T}}$ , the constant  $C_*$  and on the polynomial degree  $p$  such that the following a posteriori error estimate holds,

$$\|v - v_{h\tau}^{n,k}\|_X^2 + \|\partial_t (v - v_{h\tau}^{n,k})\|_Y^2 + \|w - w_{h\tau}^{n,k}\|_Y^2 + \|\partial_t (w - w_{h\tau}^{n,k})\|_Y^2 \leq C_{\text{rel}} (\eta^k)^2$$

with the global estimator  $\eta^k$  defined by

$$\begin{aligned} \eta^k := & \left\{ \sum_{n=1}^{N_t} \int_{I_n} \left( \sum_{K \in \mathcal{T}_h} \left( h_K^2 (\eta_{\text{R1},K}^{n,k})^2 + (\eta_{\text{R2},K}^{n,k})^2 \right) + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} h_\sigma (\eta_{\text{R3},\sigma}^{n,k})^2 \right) (t) dt \right. \\ & \left. + \sum_{K \in \mathcal{T}_h} \left( (\eta_{\text{IC},1,K}^k)^2 + (\eta_{\text{IC},2,K}^k)^2 \right) \right\}^{\frac{1}{2}}. \end{aligned} \quad (6.10)$$

*Proof.* Using (6.6) it is enough to bound the term  $\|\mathcal{R}_1(v_{h\tau}^k)\|_{X'}$ , and  $\|\mathcal{R}_2(w_{h\tau}^k)\|_Y$  to get the desired result. Let  $\varphi \in X$  such that  $\|\varphi\|_X = 1$ . First, we have

$$\begin{aligned} |\langle \mathcal{R}_1(v_{h\tau}^k), \varphi \rangle_{X',X}| &= \int_0^T \{ -(\partial_t v_{h\tau}^k + f(v_{h\tau}^k, w_{h\tau}^k), \varphi)_\Omega - (\Lambda \nabla v_{h\tau}^k, \nabla \varphi)_\Omega \} (t) dt, \\ &= \sum_{n=1}^{N_t} \int_{I_n} \{ -(\partial_t v_{h\tau}^k + f(v_{h\tau}^k, w_{h\tau}^k), \varphi)_\Omega - (\Lambda \nabla v_{h\tau}^k, \nabla \varphi)_\Omega \} (t) dt. \end{aligned}$$

Next, we employ the Green formula to have

$$(\Lambda \nabla v_{h\tau}^k, \nabla \varphi)_\Omega = \sum_{K \in \mathcal{T}_h} \{ (\Lambda \nabla v_{h\tau}^k \cdot \mathbf{n}, \varphi)_{\partial K} - (\nabla \cdot (\Lambda \nabla v_{h\tau}^k), \varphi)_K \}. \quad (6.11)$$

Using the fact that

$$\sum_{K \in \mathcal{T}_h} (\Lambda \nabla v_{h\tau}^k \cdot \mathbf{n}, \varphi)_{\partial K} = \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} ([\Lambda \nabla v_{h\tau}^k]_\sigma \cdot \mathbf{n}_\sigma, \varphi)_\sigma \quad (6.12)$$

we get

$$\begin{aligned} \langle \mathcal{R}_1(v_{h\tau}^k), \varphi \rangle_{X',X} &= \int_0^T \left\{ \sum_{K \in \mathcal{T}_h} (-\partial_t v_{h\tau}^k - f(v_{h\tau}^k, w_{h\tau}^k) + \nabla \cdot (\Lambda \nabla v_{h\tau}^k), \varphi)_K \right. \\ &\quad \left. - \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} ([\Lambda \nabla v_{h\tau}^k]_\sigma \cdot \mathbf{n}_\sigma, \varphi)_\sigma \right\} (t) dt. \end{aligned}$$

Employing the fact that  $\langle \mathcal{R}_1(v_{h\tau}^k), \varphi_h \rangle_{X',X} = 0 \quad \forall \varphi_h \in X_h^p$  and next the Cauchy–Schwarz inequality we obtain

$$\begin{aligned} \langle \mathcal{R}_1(v_{h\tau}^k), \varphi - \varphi_h \rangle_{X',X} &\leq \int_0^T \left( \sum_{K \in \mathcal{T}_h} \eta_{\text{R1},K}^{n,k}(t) \|\varphi - \varphi_h\|_K \right) (t) dt \\ &\quad + \int_0^T \left( \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} \eta_{\text{R3},\sigma}^{n,k}(t) \|\varphi - \varphi_h\|_\sigma \right) (t) dt. \end{aligned} \quad (6.13)$$

Employing first (6.13) with  $\varphi_h = \mathcal{I}_h(\varphi)$  where  $\mathcal{I}_h$  is the Clément interpolation operator introduced above and next inequalities (6.2) we have

$$\begin{aligned} \langle \mathcal{R}_1(v_{h\tau}^k), \varphi - \varphi_h \rangle_{X', X} &\leq \int_0^T \left( \sum_{K \in \mathcal{T}_h} \eta_{R_1, K}^{n, k}(t) c_A h_K \|\varphi\|_{H^1(\omega_K)} \right) (t) dt \\ &\quad + \int_0^T \left( \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} \eta_{R_3, \sigma}^{n, k}(t) c_B h_\sigma^{\frac{1}{2}} \|\varphi\|_{H^1(\omega_\sigma)} \right) (t) dt \end{aligned} \quad (6.14)$$

Setting  $C_{AB} = \max\{c_A, c_B\}$  and using the Cauchy–Schwarz inequality for sums we get

$$\begin{aligned} \langle \mathcal{R}_1(v_{h\tau}^k), \varphi - \varphi_h \rangle_{X', X} &\leq C_{AB} \int_0^T \left\{ \sum_{K \in \mathcal{T}_h} h_K^2 \left( \eta_{R_1, K}^{n, k} \right)^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} h_\sigma \left( \eta_{R_3, \sigma}^{n, k} \right)^2 \right\}^{\frac{1}{2}} \times \\ &\quad \left\{ \sum_{K \in \mathcal{T}_h} \|\varphi\|_{H^1(\omega_K)}^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} \|\varphi\|_{H^1(\omega_\sigma)}^2 \right\}^{\frac{1}{2}} (t) dt \end{aligned} \quad (6.15)$$

Next, the shape-regularity of the mesh  $\mathcal{T}_h$  implies that

$$\left\{ \sum_{K \in \mathcal{T}_h} \|\varphi\|_{H^1(\omega_K)}^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} \|\varphi\|_{H^1(\omega_\sigma)}^2 \right\}^{\frac{1}{2}} \leq \|\varphi\|_{H^1(\Omega)}$$

so that

$$\langle \mathcal{R}_1(v_{h\tau}^k), \varphi - \mathcal{I}_h(\varphi) \rangle_{X', X} \leq C_{AB} \int_0^T \left\{ \sum_{K \in \mathcal{T}_h} h_K^2 \left( \eta_{R_1, K}^{n, k} \right)^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} h_\sigma \left( \eta_{R_3, \sigma}^{n, k} \right)^2 \right\}^{\frac{1}{2}} \|\varphi\|_{H^1(\Omega)} (t) dt$$

Then, using the Cauchy–Schwarz inequality and next (6.5) we get

$$\|\mathcal{R}_1(v_{h\tau}^k)\|_{X'} \leq C_{AB} \left\{ \int_0^T \left( \sum_{K \in \mathcal{T}_h} h_K^2 \left( \eta_{R_1, K}^{n, k} \right)^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} h_\sigma \left( \eta_{R_3, \sigma}^{n, k} \right)^2 \right) (t) dt \right\}^{\frac{1}{2}} \quad (6.16)$$

Similarly, we can obtain an analogous estimate on  $\|\mathcal{R}_2(w_{h\tau}^k)\|_Y$

$$\begin{aligned} \|\mathcal{R}_2(w_{h\tau}^k)\|_Y &= \sup_{\varphi \in Y, \|\varphi\|_Y=1} |\langle \mathcal{R}_2(w_{h\tau}^k), \varphi \rangle_{Y, Y}| \\ &= \sup_{\varphi \in Y, \|\varphi\|_Y=1} \int_0^T \{ (\partial_t w_{h\tau}^k, \varphi)_\Omega - (g(v_{h\tau}^k, w_{h\tau}^k), \varphi)_\Omega \} (t) dt \\ &\leq \left\{ \sum_{n=1}^{N_t} \int_{I_n} \sum_{K \in \mathcal{T}_h} \left( \eta_{R_2, K}^{n, k} \right)^2 (t) dt \right\}^{\frac{1}{2}}. \end{aligned} \quad (6.17)$$

Employing (6.16) and (6.17) with (6.6) we get the desired estimate.  $\square$

**Remark 6.4.** The estimators of Theorem 6.3 reflect various violations of physical properties of the approximate solutions  $v_{h\tau}^k$  and  $w_{h\tau}^k$ . The estimators  $\eta_{R_1, K}^{n, k}$  and  $\eta_{R_2, K}^{n, k}$  illustrate the fact that the residual associated to (2.4) are not vanishing at the discrete level. More precisely, it shows that  $\partial_t v_{h\tau}^{n, k} - \nabla \cdot (\Lambda \nabla v_{h\tau}^{n, k}) + f(v_{h\tau}^{n, k}, w_{h\tau}^{n, k}) \neq 0$  and how far is  $\partial_t w_{h\tau}^{n, k} + g(v_{h\tau}^{n, k}, w_{h\tau}^{n, k})$  from 0. Next, the face residual estimator  $\eta_{R_3, \sigma}^{n, k}$  represents the nonconformity of the flux, i.e.  $\nabla v_{h\tau}^k \notin \mathbf{H}(\text{div}, \Omega)$ . However, at convergence ( $k \rightarrow +\infty$ ,  $h \rightarrow 0$  and  $\Delta t_n \rightarrow 0$ ) all the estimators vanish showing that our methodology is consistent.



## 6.4 Distinction of the error components

In Section 6.3, we derived an a posteriori error estimate between the exact solution and approximate solution at each time step  $1 \leq n \leq N_t$ , and at each parareal iteration  $k \geq 1$  when  $p \geq 1$ . We now provide an a posteriori error estimate distinguishing the different error components, namely the error due to the discretization in space and the error due to the parallelization in time. Note that several works also propose adaptivity of nonlinear and linear solvers thanks to heuristical distinction of the linearization error and the algebraic error. Among them we mention [28, 69, 45], see also the references therein. Recall that we are interested in solving the nonlinear problem (5.10). The parareal algorithm presented in Section 4 enables to speed up this resolution by constructing a prediction-correction procedure. In particular the parareal algorithm constructs a sequence  $(\mathbf{X}_{h,k}^{n+1})_{k \geq 1}$  that approximates the solution  $\mathbf{X}_h^{n+1}$ . We also recall that the final approximation at each time step is achieved by the accuracy of the fine solver. Besides, the high-fidelity solution is solution to the following nonlinear system of algebraic equation: find  $\mathbf{X}_h^{n+1} \in \mathbb{R}^{2m}$  such that

$$\mathbf{X}_h^{n+1} = \mathcal{F}_{\Delta T}(\mathbf{X}_h^n). \quad (6.18)$$

In the sequel, we employ an abuse of notation  $\mathcal{F}_{\Delta T}(v_h^n)$  which corresponds to the functional representation of the fine propagation  $\mathcal{F}_{\Delta T}(\mathbf{X}_h^n)$ . We consider the residual estimators of (6.7). Adding and subtracting  $\mathcal{F}_{\Delta T}(v_h^{n-1,k})$  and then using the triangle inequality, we have  $\forall t \in I_n$

$$\eta_{R_1,K}^{n,k} \leq \eta_{\text{para1},K}^{n,k} + \eta_{\text{disc1},K}^{n,k} \quad (6.19)$$

with

$$\begin{aligned} \eta_{\text{para1},K}^{n,k} &:= \left\| \frac{1}{\Delta t} (v_h^{n,k} - \mathcal{F}_{\Delta T}(v_h^{n-1,k})) \right\|_K, \\ \eta_{\text{disc1},K}^{n,k} &:= \left\| \frac{1}{\Delta t} (\mathcal{F}_{\Delta T}(v_h^{n-1,k}) - v_h^{n-1,k}) - f(v_{h\tau}^k, w_{h\tau}^k) + \nabla \cdot (\Lambda \nabla v_{h\tau}^k) \right\|_K. \end{aligned} \quad (6.20)$$

The first estimator in (6.20) is the "parareal" estimator measuring the residual of the discrete-in-time target problem (6.18). More precisely, this estimator illustrates how far the parareal solution is from the fully discrete fine problem solution. Observe that at convergence ( $k \rightarrow +\infty$ ), the estimator  $\eta_{\text{para1},K}^{n,k} \rightarrow 0$  vanishes. The second estimator in (6.20) is a discretization estimator stemming from the finite element discretization. It measures the departure of the parareal solution from the continuous spaces. Also note that when the parareal scheme converges,  $\eta_{\text{disc1},K}^{n,k} = \eta_{R_1,K}^{n,k}$ . In a same way, we have the decomposition

$$\eta_{R_2,K}^{n,k}(t) \leq \eta_{\text{para2},K}^{n,k} + \eta_{\text{disc2},K}^{n,k} \quad (6.21)$$

where

$$\begin{aligned} \eta_{\text{para2},K}^{n,k} &:= \left\| \frac{1}{\Delta t} (w_h^{n,k} - \mathcal{F}_{\Delta T}(w_h^{n-1,k})) \right\|_K, \\ \eta_{\text{disc2},K}^{n,k} &:= \left\| \frac{1}{\Delta t} (\mathcal{F}_{\Delta T}(w_h^{n-1,k}) - w_h^{n-1,k}) - g(v_{h\tau}^k, w_{h\tau}^k) \right\|_K \end{aligned} \quad (6.22)$$

so that

$$\eta_{R_2,K}^{n,k} \leq \eta_{\text{para2},K}^{n,k} + \eta_{\text{disc2},K}^{n,k}. \quad (6.23)$$

Thus, the estimator  $\eta_{\text{para2},K}^{n,k}$  is a parareal estimator for the ODE equation and the estimator  $\eta_{\text{disc2},K}^{n,k}$  is a discretization estimator representing an error induced by the finite element scheme. Note that, when the parareal scheme converges  $\eta_{\text{para2},K}^{n,k}$  vanishes and  $\eta_{\text{disc2},K}^{n,k} = \eta_{R_2,K}^{n,k}$  which shows the performance of our approach.

**Remark 6.5.** Observe that the local estimators defined above are easy to implement. For instance,

$$\begin{aligned} \left( \eta_{\text{para1},K}^{n,k} \right)^2 &= \frac{1}{\Delta t^2} \left\| v_h^{n,k} - \mathcal{F}_{\Delta T}(v_h^{n-1,k}) \right\|_K^2 \\ &= \frac{1}{\Delta t^2} \left[ \mathbf{X}_{1h}^{n,k} - \mathcal{F}_{\Delta T}(\mathbf{X}_{1h}^{n-1,k}) \right]_K^T \mathbb{M}_K \left[ \mathbf{X}_{1h}^{n,k} - \mathcal{F}_{\Delta T}(\mathbf{X}_{1h}^{n-1,k}) \right]_K \end{aligned}$$

where  $\mathbb{M}_K$  is the restriction of the mass matrix to the element  $K \in \mathcal{T}_h$ .

**Corollary 6.6.** *Let  $(v, w)$  be the solution of the continuous problem (2.1). Let  $(v_{h\tau}^k, w_{h\tau}^k)$  be the approximate solution stemming from the parareal scheme at each time step  $1 \leq n \leq N_t$  and each parareal iteration  $1 \leq k \leq \bar{k}$ . Let finally  $K \in \mathcal{T}_h$  and consider the estimators given in Theorem 6.3. Let us define the local parareal and discretization estimators by*

$$\eta_{\text{para},K}^{n,k} := h_K \eta_{\text{para}1,K}^{n,k} + \eta_{\text{para}2,K}^{n,k}, \quad \eta_{\text{disc},K}^{n,k} := h_K \eta_{\text{disc}1,K}^{n,k} + \eta_{\text{disc}2,K}^{n,k}. \quad (6.24)$$

We also define their global version as

$$\begin{aligned} \eta_{\text{para}} &:= \left\{ 2 \sum_{n=1}^{N_t} \int_{I_n} \sum_{K \in \mathcal{T}_h} \left( \eta_{\text{para},K}^{n,k} \right)^2 (t) \, dt \right\}^{\frac{1}{2}} \\ \eta_{\text{disc}} &:= \left\{ 2 \sum_{n=1}^{N_t} \int_{I_n} \left( \sum_{K \in \mathcal{T}_h} \left( \eta_{\text{disc},K}^{n,k} \right)^2 + \sum_{\sigma \in \mathcal{E}_h^{\text{int}}} h_\sigma \left( \eta_{R3,\sigma}^{n,k} \right)^2 \right) (t) \, dt \right\}^{\frac{1}{2}}. \end{aligned} \quad (6.25)$$

Then, the following a posteriori error estimate distinguishing the error components holds,

$$\begin{aligned} &\|v - v_{h\tau}^{n,k}\|_X^2 + \|\partial_t (v - v_{h\tau}^{n,k})\|_Y^2 + \|w - w_{h\tau}^{n,k}\|_Y^2 + \|\partial_t (w - w_{h\tau}^{n,k})\|_Y^2 \\ &\lesssim \eta_{\text{para}}^2 + \eta_{\text{disc}}^2 + \eta_{\text{init}}^2. \end{aligned}$$

*Proof.* The proof relies on the triangle inequality and the standard inequality  $(a+b)^2 \leq 2(a^2 + b^2) \forall (a, b) \in \mathbb{R}_+ \times \mathbb{R}_+$ .  $\square$

## 7 Numerical experiments

This section illustrates numerically our theoretical developments. The numerical results have been implemented using MATLAB. The domain  $\Omega$  is the unit square  $\Omega := [-1, 1] \times [-1, 1]$ . Concerning the spatial discretization, we use the  $\mathbb{P}_1$  finite element method (see Section 5). The time domain is  $[0, T]$  where  $T = 200$ . Concerning the coarse propagator the time step is supposed constant  $\Delta t_n = \Delta t = 0.1 \forall 1 \leq n \leq N_t$  and for the fine propagator we use a constant time step  $\delta t_n = \delta t = 10^{-3} \forall 1 \leq n \leq N^*$ . Here,  $N_t = 2000$  and  $N^* = 200000$ . The total number of elements is  $\mathcal{N}_{\text{sp}} = 5536$ . We consider the well-known phenomenological FitzHugh–Nagumo model [52, 53]. This model satisfies Assumptions 2.2 where

$$g(v, w) = -av + bw, \quad (7.1a)$$

$$f(v, w) := f_1(v) + f_2(w) \quad (7.1b)$$

with

$$f_1(v) := \lambda v(v-1)(v-\theta) \quad \text{and} \quad f_2(w) = \lambda w$$

and  $a, b, \lambda, \theta$  are given parameters satisfying  $a \geq 0$ ,  $b \geq 0$ ,  $\lambda > 0$  and  $0 < \theta < 1$ . More precisely, these parameter values are given by  $\lambda = 1$ ,  $\theta = 0.25$ ,  $a = 0.005$ , and  $b = 0.025$ . The nonlinear system (5.4) is then solved thanks to the Newton method with a stopping criterion  $\varepsilon = 10^{-12}$  (see also Remark 3.1).

In a first study, we are interested in the shape of the numerical solution after several time steps and in the behavior of the estimators at convergence of the Newton solver given by Theorem 6.3.

Figure 2 illustrates the temporal evolution of the transmembrane potential  $v_h$  and the recovery variable  $w_h$  as functions of the time step, in the case where the Newton solver has converged. The transmembrane potential  $v_h$  (red curve) represents the electrical activation of a cardiac cell, whereas the recovery variable  $w_h$  (blue curve) accounts for the delayed response of ionic channels. The overall dynamics can be divided into three main phases. First, during the *rapid depolarization* phase, the membrane potential  $v_h$  increases rapidly due to the sudden opening of sodium channels while  $w_h$  responds more slowly to this excitation. Then, in the *repolarization* phase, the variable  $w_h$  grows large enough to inhibit the dynamics of  $v_h$ , which begins to decrease. This corresponds to the closure of sodium channels and the delayed opening of potassium channels. Finally, during the *return to the resting state* phase, both  $v_h$  and  $w_h$  gradually return to their

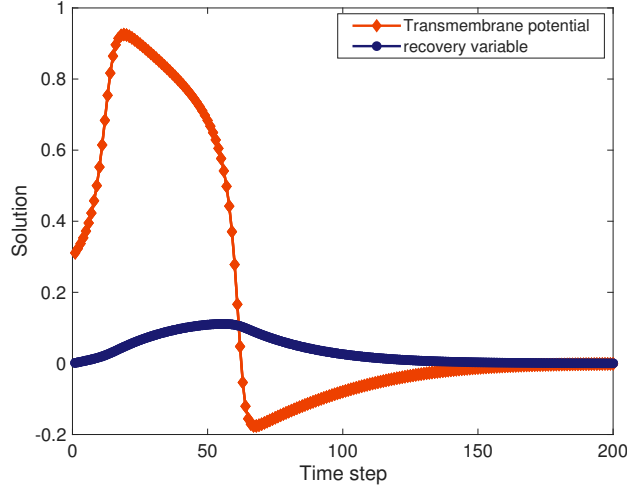


Figure 2: Temporal evolution of the high-fidelity numerical solution at a fixed spatial point, computed for 5536 mesh elements.

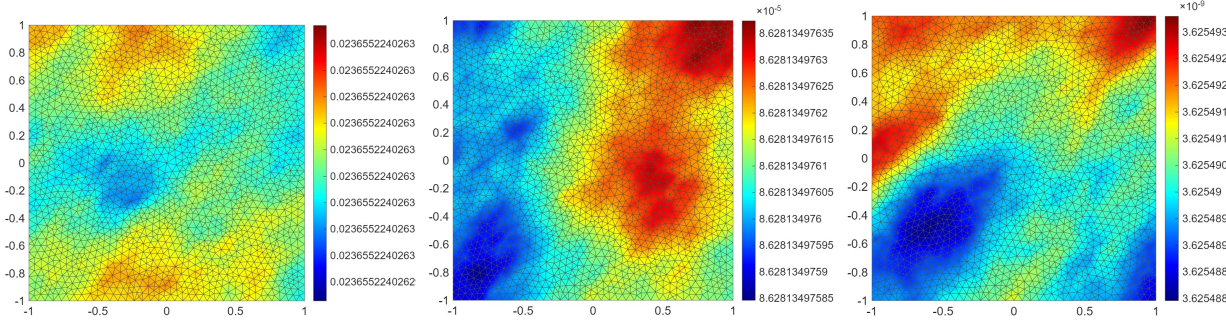


Figure 3: Distribution of the error between the high fidelity solution and the parareal solution. Error for transmembrane potential  $k = 1$  (top left),  $k = 4$  (top middle),  $k = 9$  (top right).

baseline values ( $v_h = w_h = 0$ ). The resulting trajectories highlight the coupled nature of excitation and recovery mechanisms during the signal propagation.

In Figure 3 and 4 we represent the distribution of the error between the high fidelity numerical solution and the parareal solution for several parareal iterations. We have chosen the time step  $n = 59$ . Therefore, we show  $\forall K \in \mathcal{T}_h$  the local error norms  $\|v_h^n - v_h^{n,k}\|_K$  and  $\|w_h^n - w_h^{n,k}\|_K$ . In particular, we observe that the error tends to decrease within the spatial domain  $\Omega$  when the parareal iterations increase. Table 1 is

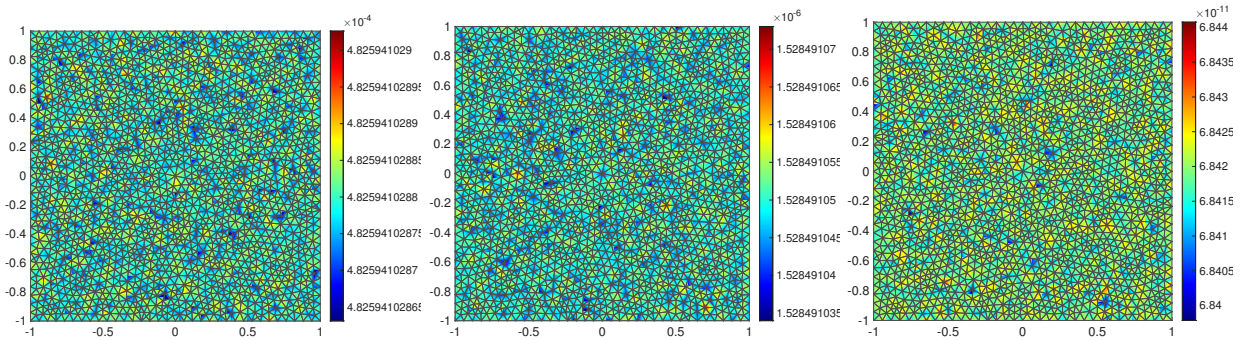


Figure 4: Distribution of the error between the high fidelity solution and the parareal solution. Error for activation variable  $k = 1$  (bottom left),  $k = 4$  (bottom middle),  $k = 9$  (bottom right).

| Para iter. | 1            | 2            | 3            | 4            | 5            | 6            | 7            | 8            | 9            |
|------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Error      | $4.73e^{-2}$ | $1.15e^{-2}$ | $1.46e^{-3}$ | $1.72e^{-4}$ | $1.84e^{-5}$ | $3.89e^{-6}$ | $8.15e^{-7}$ | $1.66e^{-7}$ | $3.34e^{-8}$ |

Table 1: Error as a function of the number of parareal iterations.

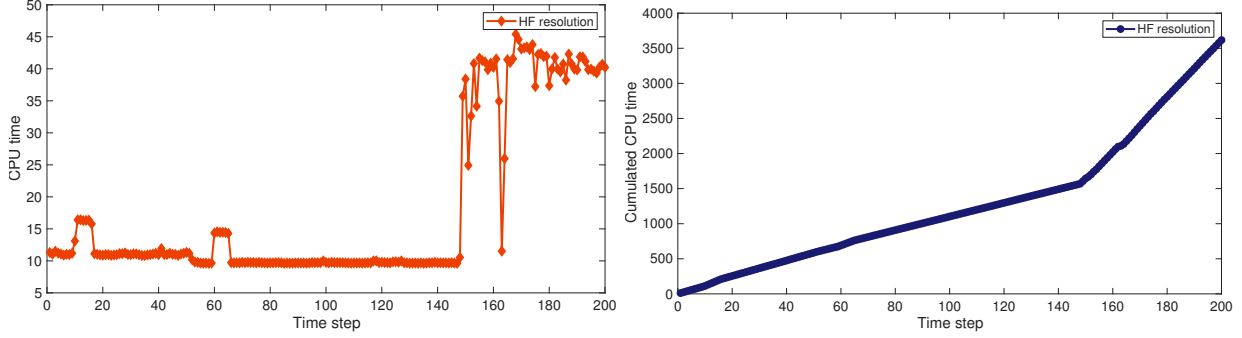


Figure 5: CPU time for sequential High-Fidelity resolution

a complement to Figure 3 . It depicts the behavior of the  $L^\infty(0, T; L^2(\Omega))$  error between the high-fidelity solution and the parareal solution for the transmembrane variable. Here, the error is defined by

$$\|v - v_{h\tau}^k\|_{L^\infty(0, T; L^2(\Omega))}^2 := \sup_{t \in [0, T]} \int_{\Omega} |(v - v_{h\tau}^k)(t, \mathbf{x})|^2 d\mathbf{x}. \quad (7.2)$$

We observe that the error decreases almost linearly and reach the desired precision in the sense of (4.3) after 9 parareal iterations. The results for the activation variable are even smaller that's why we focused on the first variable.

We displayed in Figure 5 the required CPU time to obtain the High-Fidelity numerical solution at each time step (left) and the cumulated CPU time to reach the time horizon  $T = 200$  (right). We observe from the left figure that the required CPU time is fluctuating at each time step. Overall, the cumulated CPU time for the full sequential resolution is roughly 3700 seconds.

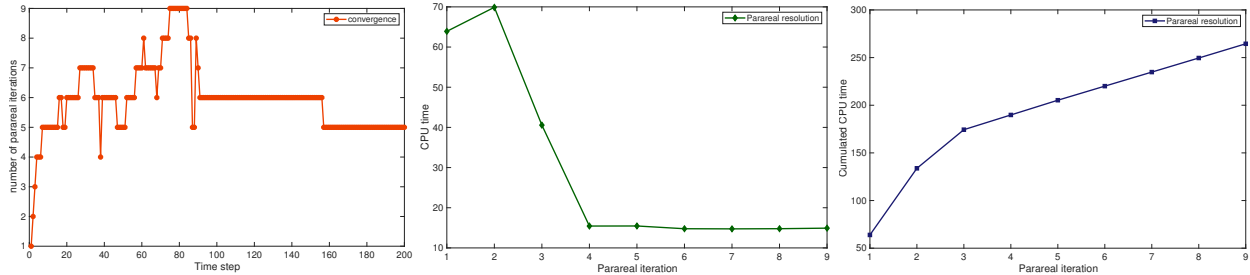


Figure 6: Number of parareal iterations to reach convergence at each time step(left), CPU time associated with the parareal resolution (middle), and cumulated CPU time associated with the parareal resolution(right)

The details of the efficiency of our parareal strategy can finally be appreciated in Figure 6. We report in the left figure the required number of parareal iterations needed to reach convergence at each time step. We observe that this number increases slowly when the time step increases. A peak occurs at the middle of the simulation before reaching roughly a constant profile around 5 – 6 iterations. We also represent the global CPU time of the simulation when the parallelization in time occurs. More precisely, as mentioned in Figure 6 a full sequential resolution requires 3700 seconds. In the parareal resolution, 200 processors are allocated (one for each time window). Our parareal resolution (as indicated on the right figure) required a total of 270 seconds to converge. For the sake of clarity we have also indicated on the left Figure the

required CPU time for each opened parareal window. Therefore, our strategy yields a gain factor in the overall CPU time of around 13.7.

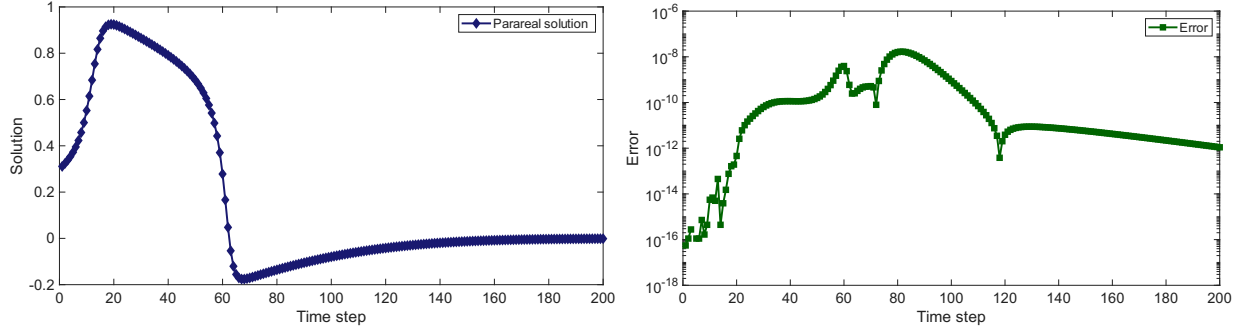


Figure 7: Parareal solution at convergence as a function of the time step (left) and error between HF solution and parareal solution (right)

Finally, in Figure 7 we represent for a fixed mesh point the evolution of the parareal solution at convergence ( $k = 9$ ) with respect to the time step (left) and its corresponding error with the HF solution (right). We observe that the behavior of the numerical solution is similar to the one provided in Figure 2. Furthermore, the right figure shows that the error between the HF solution and the parareal solution is very small at each time step. Therefrom, we deduce that our parareal resolution saves many computational time while preserving high accuracy.

Now, we carry out numerical simulations concerning the posteriori error estimates developed in Section 6.4.

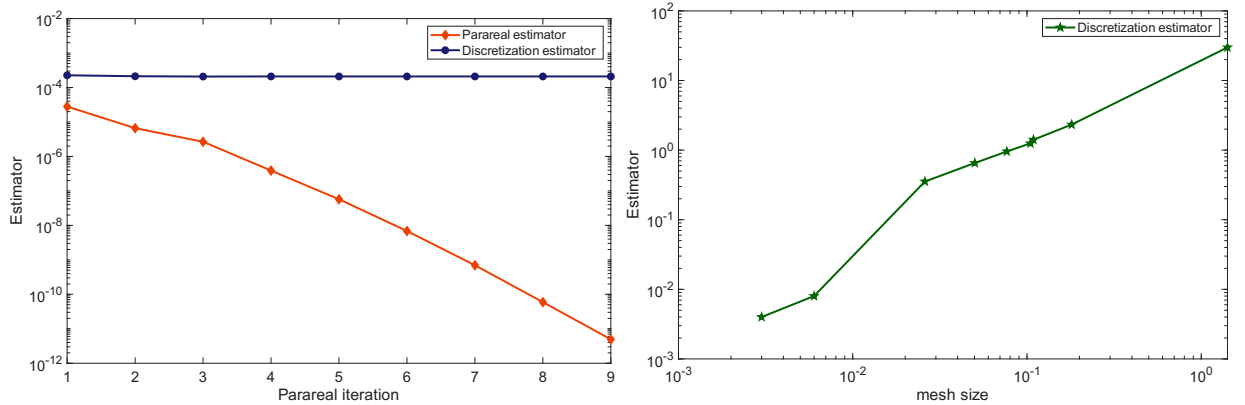


Figure 8: Estimators as a function of the parareal iteration at  $T = 200$  (left) and behavior of the global space-time discretization estimator (right)

Figure 8 (left) represents at the final simulation time  $T = 200$  the evolution of the estimators given by (6.24). More precisely, at  $n = 2000$  is represented the global parareal estimator  $\eta_{\text{para},\Omega}^{n,k} = \sum_{K \in \mathcal{T}_h} \eta_{\text{para},K}^{n,k}$  in red and the global discretization estimator  $\eta_{\text{disc},\Omega}^{n,k} = \sum_{K \in \mathcal{T}_h} \eta_{\text{disc},K}^{n,k}$  in blue as a function of the parareal iterations. We observe that the discretization estimator dominates and remains globally constant with values around  $2.5 \times 10^{-4}$  throughout the parareal iterations whereas the parareal estimator decreases until it reaches a value close to zero. Therefore, although the distinction of the error components presented in Section 6.4 is heuristic, the parareal estimator represented in red in Figure 8 exhibits a behavior that is consistent with what is typically observed for the temporal error. The right panel of Figure 8 shows the behavior of the global discretization estimator  $\eta_{\text{disc}}$  defined in (6.25). We observe a systematic decrease of  $\eta_{\text{disc}}$  as the mesh is refined. This behavior is consistent with the convergence of the spatial discretization, for which the discretization error tends to zero as  $h \rightarrow 0$ .



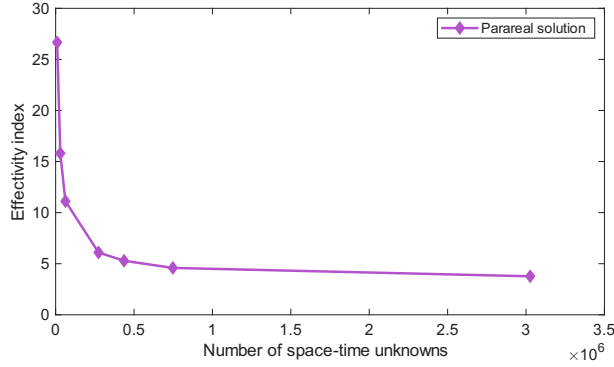


Figure 9: Effectivity index as a function of the number of space-time unknowns.

The effectivity index, denoted by  $I_{\text{eff}}$ , is defined as the ratio of the total error estimator  $\eta^k$  to the total error norm. It is reported in Figure 9 as a function of the number of space-time degrees of freedom. Since the analytical solution of system (2.1) is not available, a highly refined space-time numerical solution is used as a reference solution. As the computational effort increases, we observe that the effectivity index consistently approaches the optimal value of 1, indicating the asymptotic exactness of the proposed error estimator.

## 8 Conclusion

In this work, we introduced a parareal-in-time algorithm for accelerating a finite element approximation of the cardiac monodomain problem. The numerical results show that only a few parareal iterations are sufficient to recover the accuracy of the sequential resolution, while yielding a substantial reduction of the computational cost. These results demonstrate the efficiency and scalability of the proposed approach for the resolution of more general cardiac electrophysiology models. Furthermore, we derived an a posteriori error estimate which is valid at each parareal iterations and each time step and we distinguish the different component of the error, namely the parareal-in-time error and the discretization error. Future research will investigate the extension of this methodology to more complex models, including the bidomain and tridomain models, and to large-scale three-dimensional simulations.

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